Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	28 September 2026
Team ID	SWTID-2026-2598
Project Name	AI BlogNest API
Maximum Marks	4 Marks

Functional Requirements

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User Registration	Registration through a form (name, email, password); passwords hashed with bcrypt before storage.
FR-2	User Authentication	Login via email & password; issuance of a signed JWT (7-day expiry) on success.
FR-3	Blog Management	Create, read (all/by ID), update, and delete blog posts; only the authoring user may update or delete their own post.
FR-4	AI Content Generation	Generate a structured blog draft (title, content) from a topic; automatically infer a category when none is supplied.
FR-5	AI Content Summarization	Summarize any supplied block of text into a short, readable paragraph.
FR-6	Input Validation	Validate required fields on registration, login, and blog/AI requests using express-validator before reaching the database or AI service.
FR-7	Centralized Error Handling	Return consistent JSON error responses for validation errors, authentication failures, and unhandled exceptions.

Non-functional Requirements

FR No.	Non-Functional Requirement	Description
NFR-1	Usability	Simple, predictable JSON REST endpoints, testable directly

FR No.	Non-Functional Requirement	Description
		with Postman or Thunder Client with no additional setup.
NFR-2	Security	JWT-based stateless authentication, bcrypt password hashing, .env-based secret management, and ownership checks on write operations.
NFR-3	Reliability	Centralized error-handling middleware ensures the server returns a structured error response instead of crashing on unexpected input or failures.
NFR-4	Performance	Lightweight Express routes with non-blocking async I/O; the dominant latency is the Gemini API call (typically 2–10 seconds) rather than the database.
NFR-5	Availability	Can be deployed on any Node.js-compatible host; MongoDB Atlas can be used in place of a local database for managed uptime.
NFR-6	Scalability	Stateless JWT authentication allows multiple Express instances to run behind a load balancer; MongoDB can scale via Atlas clusters/sharding as data grows.